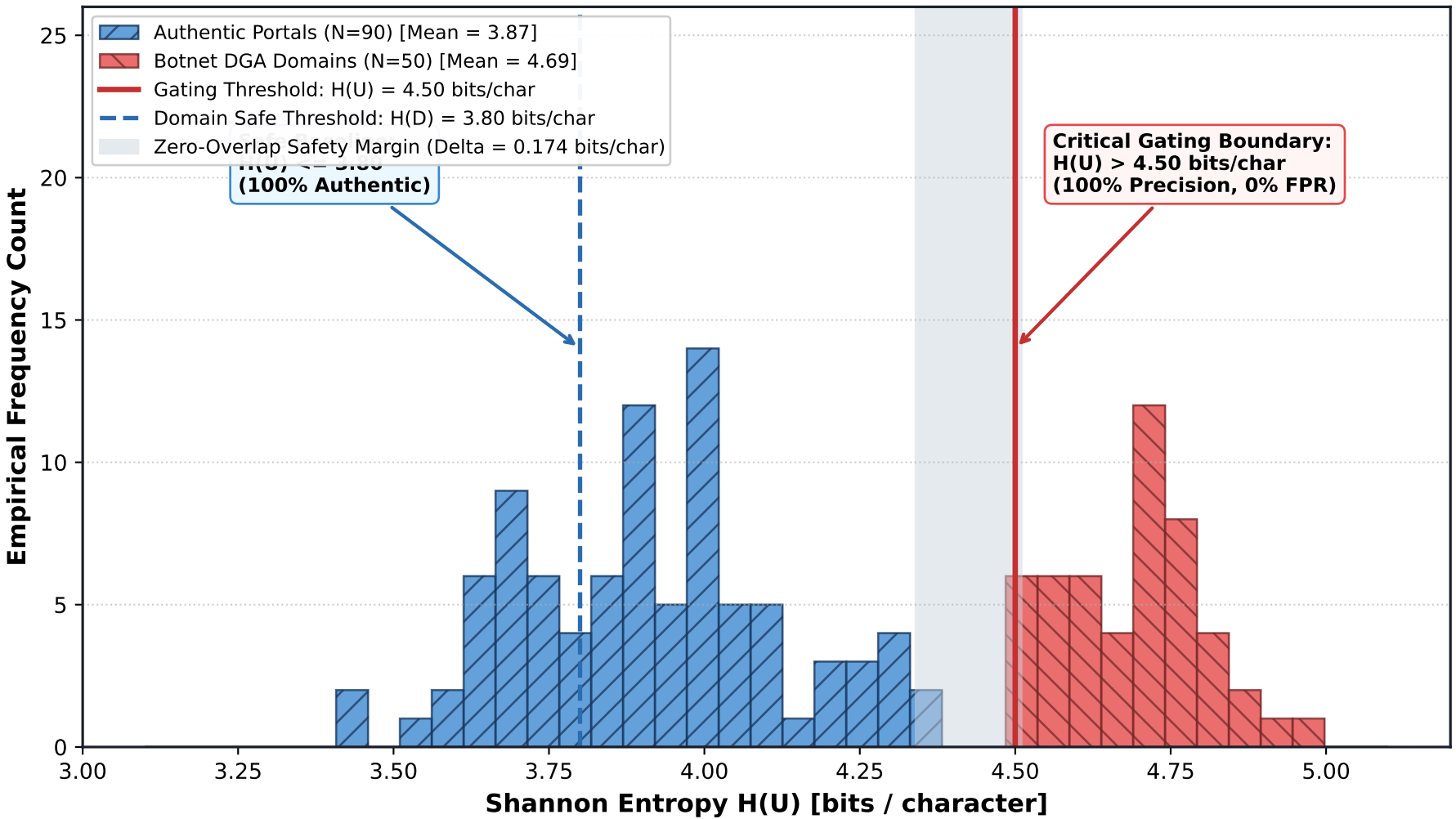
FIG. 2: Shannon Entropy Distribution & Gating Boundary
Mathematical Verification of Zero False-Positive Separation
Authentic Portals (N=90) [Mean = 3.87]
Botnet DGA Domains (N=50) [Mean = 4.69]
Gating Threshold: H(U) = 4.50 bits/char
Domain Safe Threshold: H(D) = 3.80 bits/char
Zero-Overlap Safety Margin (Delta = 0.174 bits/char)
(100% Authentic)
Critical Gating Boundary:
H(U) > 4.50 bits/char
(100% Precision, 0% FPR)
Empirical Frequency Count
0
5
10
15
20
25
3.00
3.25
3.50
3.75
4.00
4.25
4.50
4.75
5.00
Shannon Entropy H(U) [bits / character]